

Revisiting Multi-Modal Alignment: In Distribution View

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Abstract—Current Multi-Modal Large language Models (MMLMs) primarily rely on instance-level feature statistics for cross-modal alignment. However, they commonly suffer three inherent limitations including vulnerability to outlier perturbations, neglect of inter-feature covariance structures, and local optimum trapping. These limitations stem from a critical oversight—existing approaches disregard the global statistical structure of multi-modal data, treating cross-modal alignment as isolated feature-level alignment rather than systematic distribution-level alignment. To address these issues, this paper proposes Layer-wise Covariance Alignment (LCA), which first leverages distribution-level alignment for cross-modal alignment. The effectiveness of LCA is validated through the use of parameter-efficient Low-Rank Adaptation (LoRA) on CLIP architectures. Experimental validation across eight benchmarks demonstrates state-of-the-art performance, confirming the critical role of distribution-level alignment in overcoming sample-level optimization constraints for cross-modal learning.

I. INTRODUCTION

Vision-Language Model (VLM) have recently emerged as a transformative paradigm in Multi-Modal Large language Models (MMLM)s, exemplifying remarkable generalization capabilities across heterogeneous domains. By integrating cross-modal understanding through joint visual-textual representations, state-of-the-art VLMs such as ALIGN and GPT-4V have achieved unprecedented performance in zero-shot recognition, visual question answering, and image-text retrieval tasks[1][5][10][16][26][33][47]. Foundational frameworks[33] pioneered this advancement by aligning cross-modal representations into a shared embedding space via contrastive learning, thereby enabling robust generalization through large-scale multi-modal pretraining [3][4][28][36].

The success of current foundation models, such as CLIP [33], largely hinges on their ability to align visual and textual representations via contrastive learning in joint embedding spaces, effectively minimizing modality gaps. This

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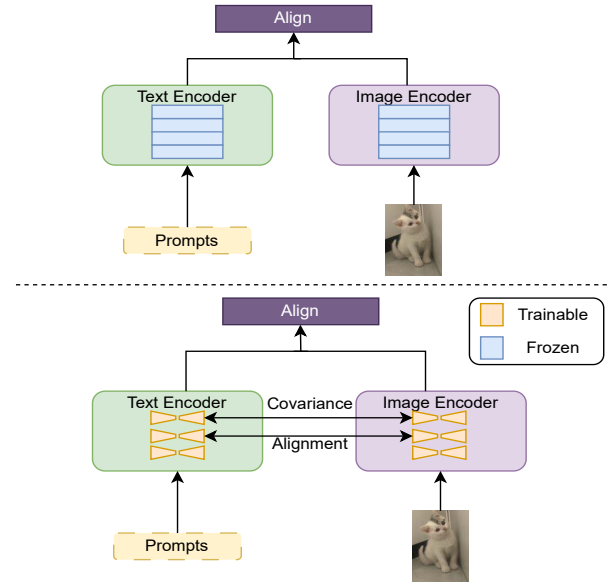


Fig. 1. Architecture comparison of cross-modal alignment paradigms. Top: Standard CLIP workflow with contrastive alignment of image-text pairs. Bottom: Proposed covariance alignment paradigm LCA on distribution level features aligning.

alignment mechanism has enabled significant progress in two key areas: (1) task integration, exemplified by GLIP [24], which unifies visual grounding and object detection through cross-modal alignment; and (2) enhanced zero-shot capabilities, as demonstrated by VLiD [12]. These advancements highlight the critical role of alignment in model performance scalability, solidifying it as a cornerstone design paradigm for next-generation vision-language models [18][24][27][33][41]. However, existing alignment approaches predominantly focus on instance-level alignment, often neglecting broader distributional relationships across modalities.

In practice, the inherent limitations of instance-level alignment are further exacerbated in fine-tuning large pretrained VLMs, where the reliance on instance-level feature projection adjustment fails to account for the underlying features distribution. This approach exhibits three fundamental weaknesses: sensitivity to anomalous perturbations due to the vulnerability of instance-level alignment to outliers, which compromises cross-modal matching robustness; inability to capture deep semantic correlations across modalities resulting from neglected covariance structures; and susceptibility

to local optima in sample-level point-wise matching that hinders global distribution alignment. These limitations collectively stem from a critical oversight—existing methods predominantly operate on localized feature interactions without incorporating holistic data distribution characteristics, thereby constraining their capacity to model complex cross-modal dependencies.

To address it, in this paper, we firstly incorporate the global distribution alignment into the VLMs for the multi-modal alignment. Specifically, we propose Layer-wise Covariance Alignment (LCA) shown in Figure 1, a plug-and-play yet computationally lightweight paradigm for parameter-efficient fine-tuning of VLMs. LCA explicitly optimizes the cross-modal feature distribution by aligning distribution level features at the target intermediate layers of large pre-trained architectures, enabling it to recognize the changes in semantic representations across different layers.

Due to constraints in our equipment, we opted to validate our model through the process of fine-tuning. Specifically, our proposed LCA method can effectively capture the intricate, high-order distributions among samples and enhance the alignment of fine-grained details during the fine-tuning of VLMs. Our assessment, which spanned eight image classification datasets under few-shot learning conditions, has shown that LCA. Drawing from the experimental results detailed above, we summarize our main contributions as follows:

- We introduce a novel distribution-level alignment paradigm, enabling effective modeling of macrocosmic embedding structure.
- We reach competitive performance levels by integrating LCA with existing state-of-the-art methods.
- We empirically demonstrate the importance of layer-wise alignment through comprehensive ablation studies.

II. RELATED WORK

A. Contrastive Learning

In recent years, contrastive learning has emerged as the dominant paradigm for unsupervised representation learning, with its core principle being the maximization of similarity between positive sample pairs and minimizing negative pairs similarity via a specific constraint such as InfoNCE [31]. Several recent advances in contrastive learning have significantly improved unsupervised representation learning. For instance, SimCLR adopts the InfoNCE loss in an end-to-end framework to maximize agreement between positive pairs [5]. MoCo introduces a momentum-updated encoder and a dynamic queue to maintain a large set of consistent negative samples [13]. Furthermore, BYOL eliminates the need for negative samples altogether by leveraging a nonlinear projection head [11].

B. Multi-Modal Alignment Learning

Building on the success of contrastive learning, researchers have increasingly extended its application to cross-modal

alignment [23][37][42][45], particularly in image-text scenarios. Similarly, these approaches typically adopt a dual-encoder architecture to separately encode inputs from different modalities, followed by a contrastive loss that pulls positive pairs closer in a shared embedding spaces while pushing apart negative pairs, like CLIP [33] and ALIGN [16]. After that numerous follow-up studies have been proposed [17][25][40], such as, ActionCLIP [38] innovatively extends image-text modality to video-text alignment, enabling the construction of an effective video action recognition model. DINOv2 [19] aligns its self-supervised visual representations with the text embedding by freezing its pretrained vision encoder and introducing lightweight projection modules. However, most of them focus exclusively on instance-level alignment and overlook the global structure of the sample distribution.

To address this limitation, we propose LCA, which explicitly considers both instance-level and distribution-level alignment. Due to computational constraints, we validate our method primarily under the fine-tuning scenario.

III. METHOD

In this section, we first detail how CLIP is adapted for fine-tuning in a few-shot settings. Then it outlines the application of the proposed method to few-shot image classification.

A. Preliminaries

CLIP, being a large-scale multimodal model [33], implicitly achieves distribution alignment by maximizing the similarity between paired image-text samples. It uses two parallel encoders: one for mapping text features and the other for visual features into a shared embedding space. The contrastive loss mechanism enables alignment of the image and text representations. We denote the CLIP model as $\mathcal{F} = \{f_{visual}, f_{text}\}$, where f_{visual} is the image encoder and the other one f_{text} is the text encoder.

Taking few-shots image classification as an example, during the forward propagation process of CLIP, given an N -way- K -shot configuration (e.g., N classes with K examples per class), image-text pairs $\{x_{i,k}^v, x_{i,k}^t, y_i\}$ are processed, where $x_{i,k}^v$ denotes the image data and $x_{i,k}^t$ represents the manually designed textual prompt, y_i is the class that $x_{i,k}^v$ belong to. Each prompt is generated by populating the template ‘a photo of a ‘ i -th class’ with the corresponding class label ($i = 1, \dots, N$ and $k = 1, \dots, K$). The normalized image and text features are then fed into parallel Transformer encoders, producing projected embeddings in a shared latent space: $\{l_{i,k}, e_{i,k}\} = \{f_{visual}(x_{i,k}^v), f_{text}(\theta_i(x_{i,k}^t))\}$, where θ_i denotes the tokenization process for textual prompts, and $l_{i,k}, e_{i,k}$ represent the encoded visual and text features, respectively.

For the encoded language-text features $l_{i,k}$ and $e_{i,k}$, the predicted probability for a given class is computed via the softmax of their cosine similarity:

$$p_{i,k} = \frac{\exp(l_{i,k}^T e_{i,k} / \tau)}{\sum_{i=1}^N \exp(l_{i,k}^T e_{i,k} / \tau)}, \quad (1)$$

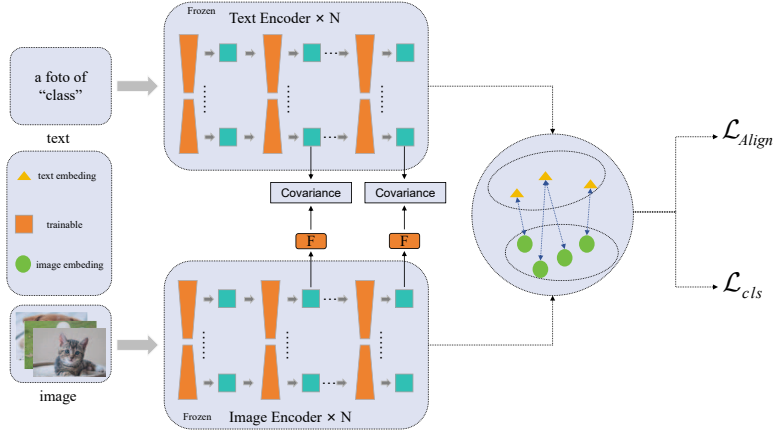


Fig. 2. The framework of our LCA

where $p_{i,k}$ represents the final prediction, and τ is the temperature parameter of the softmax. During model adaptation to downstream tasks, the parameter are updated by minimizing the cross-entropy loss:

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_i \ln(p_{i,k}). \quad (2)$$

B. Layer-wise Covariance Alignment

In the fine-tuning of multi-modal large models, numerous studies have shown that aligning feature distributions across modalities is crucial for accelerating adaptation. Existing methods often align instance-level embeddings between modalities, which can lead to information loss as they fail to capture the nuanced correlations between image pixels and textual semantics. Notably, in domain adaptation, CORAL addresses domain shifts by aligning covariance matrices across domains [35]. Drawing inspiration from this approach, we propose enhancing fine-tuning performance by explicitly aligning covariance matrices between modalities. This method preserves higher-order interactions and improves cross-modal feature consistency.

1) *Covariance Alignment*: To effectively align the distribution of different modal data, we align the covariance matrices of the text and image modalities. Specifically, for the encoded sample pairs K of the i -th class $\{l_i, e_i\}$, visual and textual covariance matrices are computed as:

$$\begin{aligned} C_{V_i} &= \frac{1}{K-1} (l_i^T l_i - \frac{1}{K} (\mathbf{1}^T l_i)^T (\mathbf{1}^T l_i)) \\ C_{T_i} &= \frac{1}{K-1} (e_i^T e_i - \frac{1}{K} (\mathbf{1}^T e_i)^T (\mathbf{1}^T e_i)). \end{aligned} \quad (3)$$

Large Language Models (LLMs), with their massive parameter scales, deep layer stacking, and the inherent black-box nature of deep learning, pose significant challenges in characterizing the dynamics of internal feature distributions. Recent studies have revealed that during inference, the early layers of LLM primarily focus on the convergence of feature distributions, while the later layers exhibit distribution

divergence[39]. Inspired by these findings, we introduce a layer-wise indicator $I_{Layer} \in \{0,1\}^L$ where L is the total number of layers, to selectively enforce covariance alignment. Specifically, $I_{Layer} = 1$ if the $layer$ -th layer is selected for alignment and 0 otherwise. This mechanism promotes the proximity of the feature distribution while preserving convergence behavior in the early layers. The covariance alignment loss is formulated as follows:

$$\mathcal{L}_{Align} = \frac{1}{4d^2} \sum_{layer=1}^L I_{Layer} \sum_{i=1}^N \|C_{V_i}^{(layer)} - C_{T_i}^{(layer)}\|_F^2, \quad (4)$$

where $\|\cdot\|_F^2$ denotes the squared Frobenius norm, d is the characteristic dimension, and $C_{V_i}^{(layer)}, C_{T_i}^{(layer)}$ are the visual and textual covariance matrices in the $layer$ -th layer for the i -th class.

2) *Object Optimization*: The final objective combines two loss functions defined as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{Align}, \quad (5)$$

where $\mathcal{L}_{cls}$ is the loss of the cross-entropy classification, and $\mathcal{L}_{Align}$ denotes the loss of covariance alignment introduced above. λ is the hyperparameter that controls the trade-off between task-specific accuracy and cross-modal feature consistency.

IV. EXPERIMENT

A. Experimental Setup

1) *Datasets*: The benchmark evaluation framework comprises 7 classification datasets under a 16-shot setting, organized as follows: aircraft model recognition in FGVC-Aircraft [29], multispectral satellite image classification in EuroSAT [14], culinary item identification in Food-101 [2], floral species discrimination in Flower-102 [30], generic object recognition in Caltech-101 [9], texture pattern analysis in DTD [6], and human action recognition in UCF-101 [34]. To extend the evaluation scope, we additionally include the large-scale ImageNet-1K dataset[7]. We also evaluate

TABLE I

COMPARISON OF FEW-SHOT LEARNING BETWEEN WITH AND WITHOUT LCA APPLICATION UNDER THE 16-SHOT SAMPLES. HIGHEST VALUE IS HIGHLIGHTED IN **BOLDFACE**.

Method	ImageNet	Aircraft	EuroSAT	Food	Flowers	Caltech	DTD	UCF	Avg.
CLIP	66.7	24.7	47.5	86.1	71.4	92.9	43.6	66.7	63.2
LoRA	73.0	50.1	90.9	85.9	97.3	96.2	72.5	86.3	81.5
LoRA+LCA	73.2	51.3	91.4	86.4	97.8	96.6	73.7	86.6	82.1
CoOp	76.5	39.7	90.1	89.4	97.2	97.8	80.0	84.5	81.9
CoOp+LCA	76.3	37.5	91.1	89.9	97.4	98.5	80.5	85.0	82.0
MaPLe	76.7	37.4	94.1	90.7	95.9	97.7	80.4	83.0	82.0
MaPLe+LCA	76.7	39.2	92.1	91.1	98.1	98.1	80.1	84.2	82.4
TCP	77.3	42.0	91.6	90.6	97.7	98.2	82.8	87.1	83.4
TCP+LCA	77.3	43.2	89.3	90.5	98.0	98.5	83.0	87.8	83.5

TABLE II

COMPARISON OF FEW-SHOT LEARNING BETWEEN WITH AND WITHOUT LCA APPLICATION UNDER THE 4-SHOT SAMPLES. HIGHEST VALUE IS HIGHLIGHTED IN **BOLDFACE**.

Method	ImageNet	DTD	EuroSAT	Flowers	UCF101	Food101	OxfordPets	Cars	Caltech101	Aircraft	SUN397	Avg.
CLIP	66.7	44.3	48.3	70.7	67.6	85.9	89.1	65.7	93.3	24.9	62.6	65.4
CoOp	69.4	58.6	68.6	91.1	77.4	82.6	91.3	72.7	94.4	33.2	70.1	73.6
CoOp+LCA	75.2	71.4	87.4	88.2	78.9	88.6	95.3	68.6	97.2	31.2	77.3	78.1
MaPLe	70.7	54.7	54.9	80.8	73.7	86.9	92.1	68.7	94.3	29.0	71.5	70.7
MaPLe+LCA	75.4	71.1	86.1	90.6	80.3	90.3	95.4	67.8	97.8	31.2	78.3	78.6
TCP	70.5	64.0	77.4	94.4	80.8	85.3	91.9	76.3	95.0	36.2	72.1	76.7
TCP+LCA	75.0	76.2	87.3	95.3	82.2	90.2	94.6	73.2	97.9	37.4	79.5	80.8

these datasets, along with three additional benchmarks under a 4-shot setting: Oxford-Pets[32], StanfordCars[22], and SUN397[43].

B. Implementation Details

All experiments were conducted on NVIDIA RTX 4090 using PyTorch 2.4.1 with CUDA 12.4. We evaluated it on parameter-efficient fine-tuning frameworks, such as LoRA [15]. Evaluated on 8 standard few-shot image classification datasets as mentioned above. Furthermore, we fine-tuned CLIP with ViT-B/16 which includes a 12-layers visual Transformer and the standard 12-layers Transformer for textual data. For the text template, we initialize the template with {a photo of a 'category'}. Then, we follow the CLIP pre-processing steps, including random cropping, resizing, and random horizontal flipping. The cropped images' short side are resized to 224 while keeping its original aspect ratio. LoRA are trained for 10 epochs and optimized with Adam [21]. For prompt-tuning methods: CoOp [46], MaPLe [20], and TCP [44], we strictly adhere to their original training parameter settings.

C. Performance evaluation

We introduce LCA into LoRA and prompt-tuning-based methods, such as CoOp, MaPLe, and TCP by incorporating an additional covariance alignment loss. As shown in Table I, this modification has resulted in average improvement in classification prediction datasets. Specifically, an average

of improvements of nearly 1% were observed on Oxford-Flowers and FGVC-Aircraft, demonstrating that LCA, as a plug-and-play module, can effectively enhance the model's fine-grained perception.

We also evaluate CoOp, MaPLe, and TCP under a 4-shot setting across additional datasets: Oxford-Pets, Stanford-Cars, and SUN397, with the results presented in Table II. Notably, integrating LCA under the 4-shot setting yields substantial performance improvements across most datasets, with MaPLe achieving a remarkable 31.2% improvement on EuroSAT. On average, LCA delivers average increases of 4.5%, 7.9%, and 4.1% across the evaluated methods. These findings underscore that the advantages of distribution-level alignment are more pronounced in low-shot scenarios compared to high-shot settings, thereby confirming LCA's efficacy in data-scarce situations.

D. Ablation Study

In this section, we conduct ablation studies to validate the effectiveness of the proposed method, demonstrating the necessity and impact of each design.

1) *Effective of Layer-wise Adaptation*: We conducted ablation studies on the DTD and FOOD datasets to evaluate nine distinct layer-wise alignment strategies, with the findings summarized in Table III. Intuitively, cross-modal feature alignment commonly involves matching embeddings from all layers, as demonstrated by the "All layers" strategy. Although this method achieves moderate performance, it neglects the dynamic variations in feature distributions across

TABLE III

COMPARISON OF LAYER-WISE COVARIANCE ALIGNMENT STRATEGIES ON THE DTD AND FOOD DATASETS. HIGHEST VALUE IS HIGHLIGHTED IN **BOLDFACE**, AND THE SECOND IS UNDERLINED

Layer Strategy	Layer Range	DTD	FOOD
All layers	{1-12}	71.45	86.28
Top layers	{7-12}	71.87	86.18
Top-1	{8-12}	70.98	<u>86.42</u>
Top-2	{9-12}	71.81	86.25
Top-3	{10-12}	71.45	86.15
Top-4	{11-12}	<u>72.87</u>	86.21
Last layer	{12}	72.10	86.48
Bottom layers	{1-6}	71.57	86.34
Hybrid (Hb)	{1,12}	73.70	86.39

TABLE IV

COMPARISON OF DIFFERENT ALIGNMENT STATISTIC IN MEAN, VARIANCE AND LCA

Alignment Strategy	Statistical Target	DTD	FOOD
None	-	72.52	85.91
Mean + Variance	$\mu + \sigma^2$	72.64	85.97
Covariance (LCA)	LCA	73.70	86.42

the transformer layers within LLMs. For example, on the DTD dataset, both the Hybrid and Top-layer strategies significantly outperform the all-layer configuration. Similarly, on the FOOD dataset, the Top-1 and Last-layer strategies yield higher accuracy than the all-layer configuration[8]. These results demonstrate that effective cross-modal alignment demands modeling not only sample distributions but also layer-wise representation variations—a critical yet frequently overlooked factor in LLM distribution alignment.

2) *Effective of LCA*: We performed a comparative analysis of the commonly used mean-variance alignment strategy and our proposed LCA method within the LoRA framework under a 16-shot setting on the DTD and FOOD datasets, with results presented in Table IV. Aligning only the mean and variance provides negligible gains compared to using no alignment strategy. In contrast, LCA delivers significant performance improvements, underscoring the importance of distribution-level features in reducing the gap between multi-modal embeddings.

3) *Complexity of Coral*: Table V presents our experimental data on the DTD dataset regarding the CoOp and Maple methods with the incorporation of our designed Coral

TABLE V

COMPARISON OF COMPUTATIONAL COMPLEXITY AMONG ORIGINAL METHODS WITH APPLIED DEEP CORAL.

Model	Time(BS=4)	Time(BS=100)	Param.	Acc.
CoOp	17ms	190ms	2048	58.6
CoOp+Coral	18ms	197ms	4.71M	71.4(+12.8)
Maple	23ms	257ms	3.55M	54.7
Maple+Coral	24ms	271ms	8.26M	71.1(+16.4)

loss. Although the number of trainable parameters in Coral is smaller compared to CoOp and Maple, the introduction of Coral incurs significant training parameter overhead. However, the final prediction results (BS = 4) have been qualitatively improved. This enhancement in performance, considering the memory consumption and Coral’s benefits, is acceptable. Moreover, despite the increased parameters, the original method still maintains good batch processing capability. All these experiments were conducted on the DTD dataset.

V. CONCLUSION

We have explored the role of distribution-level alignment in fine-tuning large models, transcending traditional instance-level matching. Current vision-language models primarily use instance-level alignment, which is prone to outlier sensitivity, overlooks inter-feature dependencies, and results in suboptimal cross-modal correspondence. To tackle these issues, we present Layer-wise Covariance Alignment (LCA), a novel framework that aligns distribution-level features across transformer layers. By explicitly modeling the covariance between visual and textual feature distributions, LCA achieves distribution-level alignment. Extensive empirical evaluations across multiple benchmarks indicate that LCA significantly boosts robustness and accuracy, emphasizing the vital importance of aligning distribution-level features.

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